



The KuiSCIMA Dataset for Optical Music Recognition of Ancient Chinese Suzipu Notation

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Abstract. In recent years, the development of Optical Music Recognition (OMR) has progressed significantly. However, music cultures with smaller communities have only recently been considered in this process. This results in a lack of adequate ground truth datasets needed for the development and benchmarking of OMR systems. In this work, the KuiSCIMA (Jiang Kui Score Images for Musicological Analysis) dataset is introduced. KuiSCIMA is the first machine-readable dataset of the *suzipu* notations in Jiang Kui's collection *Baishidaoren Geku* from 1202. Collected from five different woodblock print editions, the dataset contains 21797 manually annotated instances on 153 pages in total, from which 14500 are text character annotations, and 7297 are *suzipu* notation symbols. The dataset comes with an open-source tool which allows editing, visualizing, and exporting the contents of the dataset files. In total, this contribution promotes the preservation and understanding of cultural heritage through digitization.

Keywords: Ancient Chinese music · Optical Music Recognition · Suzipu · Banzipu · Jiang Kui

1 Introduction

Optical Music Recognition is a discipline belonging to document analysis. Its goal is to automatically recognize symbolic musical data from documents provided as image data. In recent years, its development has thrived, and many datasets for OMR applications have been designed, the majority of which feature collections of Western music. Even though there are some OMR algorithms which have been created for Chinese music, publicly available OMR datasets are not available, thus, comparing new algorithms against the state of the art is difficult.

Therefore, in this paper the dataset KuiSCIMA (Jiang Kui Score Images for Musicological Analysis) is presented. Not only is it the first OMR dataset of *suzipu* 俗字谱 notation, it is also the first publicly available OMR dataset for Chinese music. The dataset focuses on the 17 melodized *ci* 词 lyrics using *suzipu* notation (also called *banzipu* 半字谱) from Jiang Kui's collection *Baishidaoren*

The original version of this chapter was previously published without open access. A correction to this chapter is available at https://doi.org/10.1007/978-3-031-70552-6_27

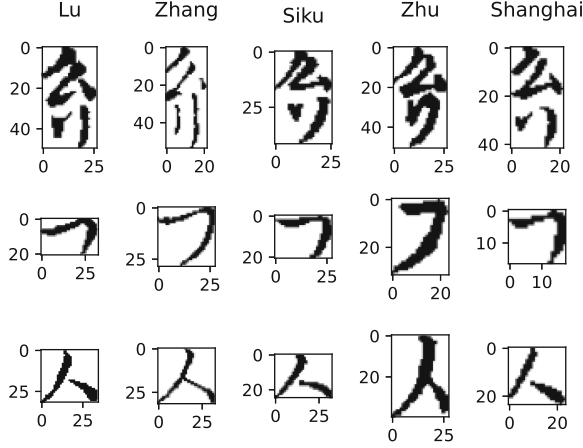


Fig. 1. Here, the first three notations of the piece *Geximeiling* 鬲溪梅令 are visualized for each edition, highlighting the differences in writing styles.

Gegu 白石道人歌曲, which was originally published in 1202. To ensure a variety of writing styles in this rather small set of pieces, five different woodblock print editions have been included: The Lu edition from 1743 [24], the Zhang edition from 1749 [25], the *Siku Quanshu* edition from the end of the 18th century [27], the Zhu edition from 1913 [23], and the *Shanghai MS* (where *MS* is short for *manuscript*) from around 1736 [26]. KuiSCIMA features symbol-level content and position annotations for both text and notation characters together with the binarized source images. A small sample of the editions is found in Fig. 1.

Baishidaoren Gegu has been chosen as the base for this dataset because Jiang Kui (1155–1221) is one of the most influential poets of Song dynasty [10], and his *ci* poems are held in high esteem. This literary genre originated as a sung form of poetry, but became mostly literary over time [21]. The collection *Baishidaoren Gegu* is therefore one of the earliest examples of melodized lyrics in China and the largest historical collection of *suzipu* notation which have been preserved until today [19]. In addition, a similar notation is used in the folk music tradition of *Xi'an Guyue* 西安鼓乐 music, which is still practiced today [6].

The KuiSCIMA dataset and the experiments in this paper promote the preservation and understanding of cultural heritage through digitization.

2 Related Works

The related works to this paper can roughly be split into three groups: The first group encompasses digital ground truth OMR datasets, the second group deals with the development of OMR algorithms, and the third group is datasets of Chinese historical documents for OCR purposes.

The first group encompasses many works, therefore, only three datasets are chosen as representatives here: [7] contains the CVC-MUSCIMA¹ dataset, which consists of handwritten music scores and is specifically designed for score writer identification and staff removal tasks. The MUSCIMA++ dataset [8] is based on CVC-MUSCIMA and provides additional annotations for individual musical symbols and the relationships between them. For Gregorian chants, there is the GregoBase dataset [1] featuring the scores in GABC notation linked to the original image sources.

In the second group, there is more cultural diversity than in the first group. For example, in [3], the authors develop a deep neural network for the transcription of handwritten mensural notation. In [12], a holistic end-to-end approach for mensural notation is developed, which is usable in settings where page content annotations are given without specific symbol positions.

Targeting OMR for Chinese traditional music, two works are chosen as representatives: [4] is a comparison of different algorithms for recognition of *gongche* notation of *Kunqu* musical theater, while [15] focuses on autonomous extraction methods of *jianzipu* notation for the instrument *guqin* 古琴. Unfortunately, even though these are papers dealing with OMR methods for Chinese music notations, they do not provide a publicly available ground truth dataset which could be used for performance comparison of new OMR algorithms.

The third group contains databases such as the Tripitaka Koreana in Han (TKH) and the Multiple Tripitaka in Han (MTH) datasets from [20], containing annotated character-level bounding boxes and text line annotations. These have been expanded to the MTHv2 dataset in [11], containing 1081663 character instances on 3199 pages in total. In [14], the ICDAR 2019 Historical Document Reading Challenge on Large Structured Chinese Family Records dataset is introduced with textline annotations on 12850 pages. The SCUT-CAB database [5] contains detailed layout annotations of 4000 pages. KuiSCIMA however has a much smaller dataset size, and only the portions containing the 17 *suzipu* pieces are annotated with character-level segmentation boxes currently.

3 Methodology

3.1 Digital Representation Format

The format used for the representation is an extension of the JSON format 1.0 introduced in [13] and thus given the version number 2.0.

The file `json_schema_suzipu.json`² is a JSON schema containing a technical documentation of the digital representation format used in this paper, and it can also be used to verify that a dataset file is of valid structure. In Fig. 2, the structure of a *suzipu* piece is visualized.

¹ The abbreviation MUSCIMA stands for MUsic SCore IMAGes, being the inspiration for the name of the KuiSCIMA dataset.

² https://github.com/SuziAI/gui-tools/blob/main/json_schema_suzipu.json (tagged v2.0).

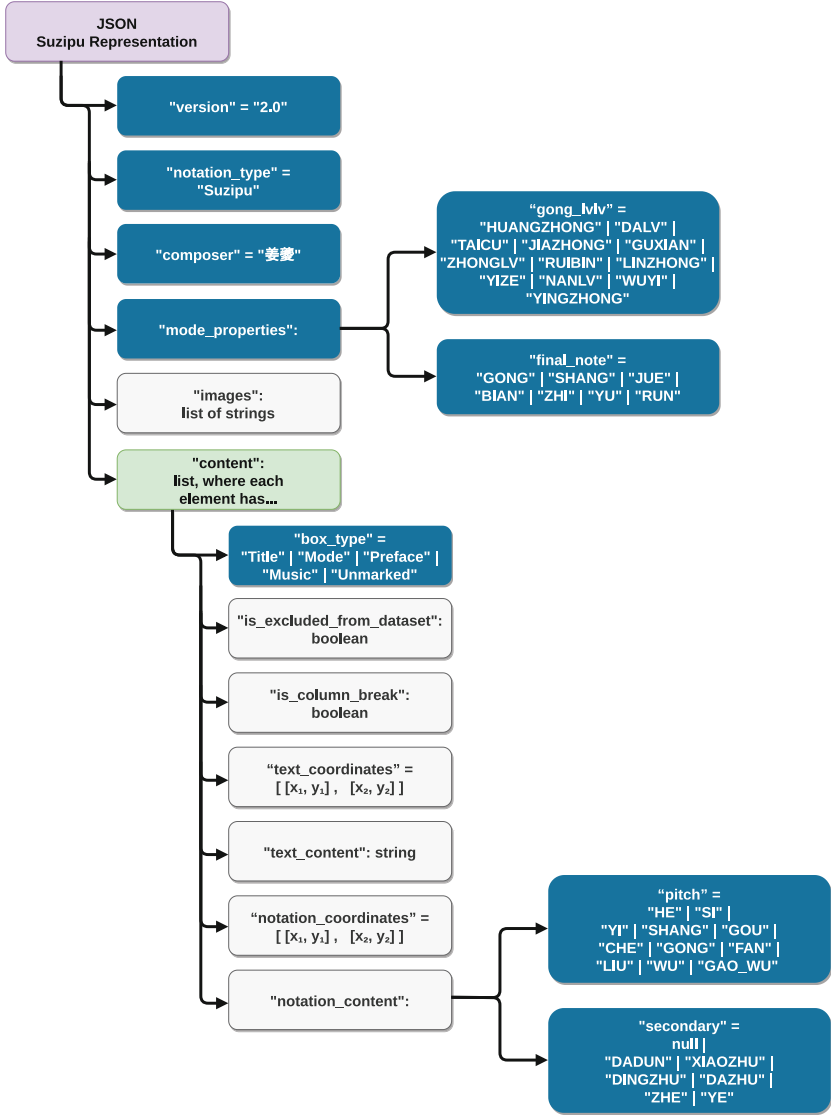


Fig. 2. Structure of the *suzipu* piece format. Blue boxes mark required fields, and grey boxes mark optional fields. (Color figure online)

For the metadata, in the root level, the following piece properties have changed in version 2.0:

1. **mode_properties**: Here, the properties uniquely defining the mode are stored, i.e., the *lǜlǜ* 律吕 which is corresponding to melodic function *gong* 宫 (stored in *gong_lv*), and the mode's final note (stored in *final_note*). As opposed

to version 1.0, now the names of the 11 *lǚlǚ* and of the 7 melodic functions are given as uppercase pinyin string, as demonstrated in Fig. 2.

2. **content**: This is an ordered list of all the individual boxes in the piece. The optional property **notation_content** for storing the notation information has changed. Instead of a plain string, it now consists of two subfields **pitch** for the pitch and **secondary** for the secondary *suzipu* symbol. The correspondences between the notation and the field representation are given in Table 1.

An introduction to the structure of the pieces of *Baishidaoren Gequ* and further explanation on the format and field choices are found in [13].

Table 1. The correspondence between the *suzipu*, its *gongchepu* name, and the JSON field representation.

<i>Suzipu</i> (Pitch)	Name Field Representation	
ム	合	"HE"
マ	四	"SI"
一	一	"YI"
么	上	"SHANG"
ㄥ	勾	"GOU"
ハ	尺	"CHE"
フ	工	"GONG"
リ	凡	"FAN"
ス	六	"LIU"
ウ	五	"WU"
㍻	高五	"GAO_WU"
<i>Suzipu</i> (Secondary)	Name Field Representation	
㍻	大顿	"DA_DUN"
リ	小住	"XIAO_ZHU"
フ	丁住	"DING_ZHU"
カ	大住	"DA_ZHU"
ㄣ	折	"ZHE"
ㄩ	拽	"YE"

3.2 Image Preprocessing

The 1743 Lu edition [24] and the *Siku Quanshu* edition [27] are available online on the Chinese Text Project [16] and have been downloaded manually. The

images are in PNG format, already binarized, and resized to a width of 800 pixel and variable height. In the *Siku Quanshu* edition of the piece *Nishangzhongxu Diyi* 霓裳中序第一, the first four columns of the piece, which contain the title and a part of the preface, show artifacts due to a suboptimal digitization of a red stamp.

The other three editions must be preprocessed to be used for the dataset:

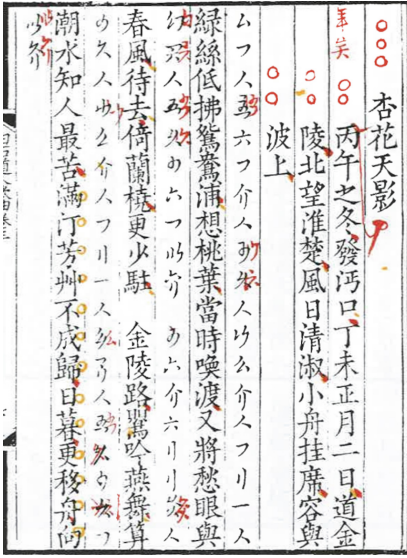
1. The 1913 Zhu edition [23] is scanned, each of the double-page images is split into halves, rotated to fit a rectangle frame if too skewed, cropped to the area of the woodblock print, converted to grayscale, binarized using OpenCV-Python’s implementation of Otsu’s method, resized to 800 pixels in width using the `cv2.INTER_AREA` flag, and then binary thresholding with a threshold of 235 is applied. Since there is still a lot of noise present, using OpenCV’s [2] `connectedComponentsWithStats` function, all connected components with an area of smaller or equal than 10 are removed.
2. For the Shanghai MS and the 1749 Zhang edition [25], a slightly different method is used: The individual pages are scanned, warped using the corrective perspective tool in GIMP if too skewed, and cropped. Since the Zhang edition contains remarks in black, yellow and red, the yellow and red ones are removed before the grayscale conversion by extracting and replacing the colored parts in HSV space. This color removal is not done for the Shanghai MS, since no colored remarks are present. Then, the images are resized to 1600 pixels in width, Otsu’s method is applied, and the images are individually split into halves. The individual stages of preprocessing are illustrated in Fig. 3.

3.3 Annotation Process

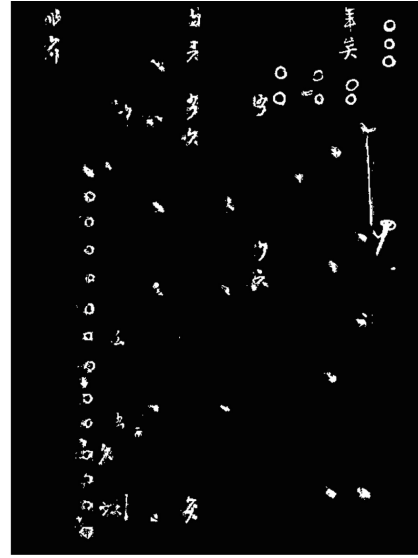
Using the *Suzipu Musical Notation Annotation Tool* introduced in [13], for each of the 17 *ci* pieces, the preprocessed pages containing the piece are selected. This includes the title, possibly the mode information and the preface, and the lyrics with the *suzipu* notation.

Then, using the HRCenterNet algorithm [18], for each page a segmentation of the woodblock print is performed individually, yielding a set of rectangles, each of which are supposed to correspond to the bounding box of a character in the image. Using this preliminary segmentation, the boxes not belonging to the piece are removed, and the boxes are checked individually and corrected if necessary, so that each bounding box encompasses the character adequately. Both text boxes and music boxes are created this way.

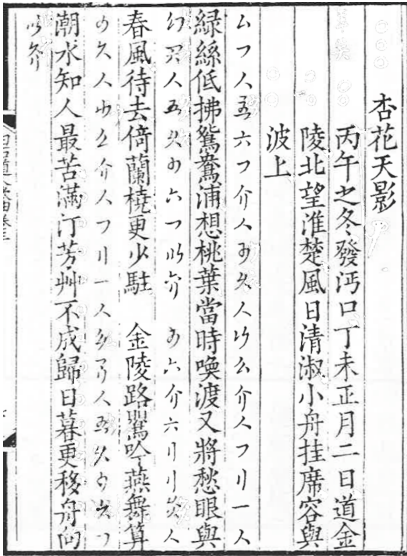
Additionally, the blank spaces in both the lyrics and music columns separating the stanzas are represented by a bounding box containing an empty string as annotation, in order to encode the typographical layout of each edition as closely as possible.



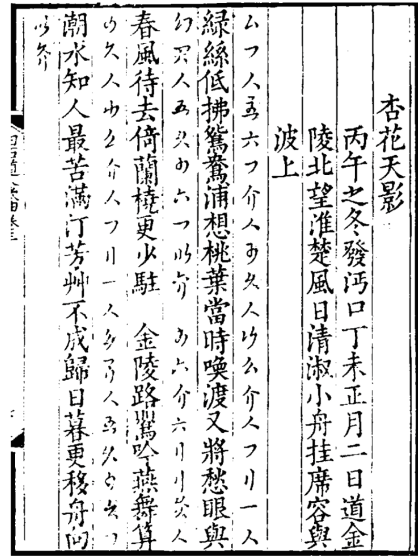
(a) The warped and cropped page: as usual in historical Chinese documents, read column-wise from right to left, each column from top to bottom



(b) Masking the pixels in HSV space: saturation between [35, 255], and value between [155, 235]



(c) Binary OR operation between the original image and the masked pixels, leading to a removal of the colors from the image



(d) Result after applying Otsu's method

Fig. 3. Visualisation of the individual preprocessing steps for images of the Zhang edition, as example the page containing *Xinghuatianting* 杏花天影 is chosen.

In the next step, each box is assigned a label according to its function in the document: E.g., boxes containing a character of the piece’s title are marked with the **Title** label, the *suzipu* characters with the **Music** label etc.

After this, each box is annotated such that the annotation corresponds with the semantic meaning of the character. In case of the text-based boxes, each annotation is a string containing the ground truth of the Chinese character. In case of the music-based boxes, each annotation is the digital representation of a single *suzipu* character, i.e., a JSON object consisting of two fields **pitch** and **secondary**, with values as described in Table 1.

Finally, the mode information is annotated and the piece is converted to the JSON representation. As a final step, all boxes in the dataset are resized so that the segmentation box is minimal, i.e., the least amount of white space is present.

3.4 Annotation Remarks

There are three reasons why the ground truth annotations of both the Chinese text characters and the *suzipu* notation symbols are not always clear:

1. All the editions in this dataset are results of woodblock printing, a process involving manually carving out the shapes of each individual character. Similar to modern prints, errors when copying the source material may occur, and errors already included in the source material itself may be copied.
2. When printing, some parts of the woodblock may already be worn or the ink is not distributed evenly, causing artifacts in the final print [17].
3. *Baishidaoren Geku* had been forgotten for over 300 years, before a manuscript appeared in the beginning of the 18th century. All editions included in this dataset thus have been copied from this manuscript in Qing dynasty, at a time where the *suzipu* notation was already uncommon and *ci* poetry had long turned from a musical practice into a mostly literary genre [21].

These factors may result in anomalous shapes or misprints of the text or the *suzipu* notation characters, such that the meaning of certain musical symbols is unclear and needs further interpretation. Some of these instances are common to all or multiple editions in the KuiSCIMA dataset, while others are restricted to individual editions.

The annotations in this dataset try to preserve the original shapes of the characters as closely as possible, while still being interpreted in the *suzipu* framework given in [13]. In case of seldom used Chinese characters, the most common variant character is used instead. For ambiguous Chinese characters, the character is annotated as described in [21]. For unclear notation symbols, case-by-case decisions had to be made which are listed in the file `annotation_remarks.pdf`.³

³ https://github.com/SuziAI/KuiSCIMA/blob/main/annotation_remarks.pdf (tagged v1.0).

3.5 Dataset Contents

KuiSCIMA contains all 17 *suzipu* pieces from *Baishidaoren Gequ* in four different editions. The pieces are listed in Table 2.

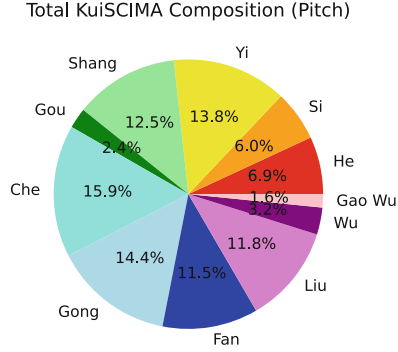
Table 2. Listing of all the 17 pieces with *suzipu* notation. Column *Index* contains the index with respect to the order of all pieces in *Baishidaoren Gequ* as given in [21].

Name	Romanization	Index
鬲溪梅令	<i>Geximeiling</i>	30
杏花天影	<i>Xinghuatianying</i>	48
醉吟商小品	<i>Zuiyinshangxiaopin</i>	49
玉梅令	<i>Yumeiling</i>	50
霓裳中序第一	<i>Nishangzhongxu Diyi</i>	59
揚州慢	<i>Yangzhouman</i>	79
長亭怨慢	<i>Changtingyuanman</i>	80
淡 $\boxplus$ 柳	<i>Danhuanliu</i>	81
石湖仙	<i>Shihuxian</i>	82
暗香	<i>Anxiang</i>	83
疏影	<i>Shuying</i>	84
惜紅衣	<i>Xihongyi</i>	85
角招	<i>Jueshao</i>	86
$\boxplus$ 招	<i>Zhishao</i>	87
秋宵吟	<i>Qiuxiaoyin</i>	88
$\boxplus\boxplus$ 犯	<i>Qiliangfan</i>	89
翠樓吟	<i>Cuilouyin</i>	90

Optical Symbolic Dataset. The optical symbolic dataset contains symbolic text and notation information with bounding boxes annotating images of the historical editions. This is the core of KuiSCIMA and the basis for the development and evaluation of OMR methods acting on *suzipu* notation.

The folder `KuiSCIMA/optical_symbolic_dataset` contains all individual editions in folders named after the corresponding editions, i.e., `01_lu.edition`, `02_zhang.edition`, `03_siku_quanshu.edition`, `04_zhu.edition`, and finally `05_shanghai_manuscript`. Inside each of these folders are two subfolders, one for the images called `images`, and one for the JSON files called `json`.

The instances in the optical symbolic dataset feature a highly imbalanced class distribution as visualized in Fig. 4. In addition, the task of identifying the composite notations is difficult, since the individual strokes of a symbol may be separated or connected. As opposed to Chinese character OCR, any *suzipu* symbol has at most two components. However, the degree of ambiguity in *suzipu* is much higher due to the small set of components and their similarity to each other. This ambiguity is shown in Fig. 5.



(a) Total pitch distribution, indicating that the pitches are highly imbalanced

Composite KuiSCIMA Composition

He	15	4	42	0	5	10
Si	33	5	43	6	8	5
Yi	137	39	50	0	146	38
Shang	49	48	93	8	16	54
Gou	4	5	17	0	3	11
Che	76	36	100	5	17	61
Gong	36	8	110	5	18	28
Fan	25	30	29	0	141	77
Liu	44	49	61	0	0	86
Wu	0	2	22	0	0	17
Gao Wu	0	3	8	5	0	3
	Dadun	Xiaozhu	Dingzhu	Dazhu	Zhe	Ye

(b) Composite notation distribution, illustrating that the pitch/secondary instances are concentrated locally

Fig. 4. The balance of KuiSCIMA dataset. In (a), the total pitch distribution of all instances is shown, while in (b), the distribution of the composite notations is illustrated.

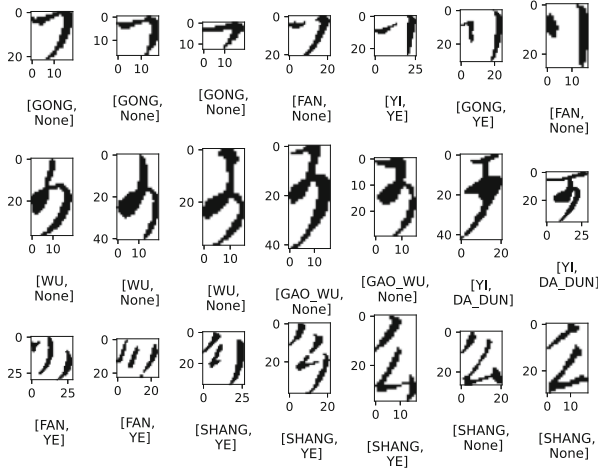


Fig. 5. In each row, examples of similar-looking notations in the Shanghai MS are visualized.

Symbolic Dataset. In addition, a purely symbolic dataset is provided, containing two editions of the 17 pieces without image data annotations.

The edition of Wu Santu [22] is the basis for the symbolic-only dataset regarding the notation and layout. The punctuation marks and the lyrics are taken from Yang Yuanzheng’s new critical edition of *Baishidaoren Gequ* in [21]. This dataset follows the historical editions of *Baishidaoren Gequ* closely, including pitch symbols which lie outside the pieces’ modes. It is found in the folder `KuiSCIMA/symbolic_dataset/wu_edition`.

For certain applications such as music generation, it is beneficial to normalize the pitches such that only pitches which are in accordance with the pieces’ modes are present in the data. This dataset, called *normalized edition* is provided in the folder `KuiSCIMA/symbolic_dataset/normalized_edition`.

4 Baseline Experiments

Since the main goal of the KuiSCIMA dataset is the use for developing and benchmarking *suzipu* OMR systems, a first baseline on the task of symbol classification shall be established in this section. The source code is open-source and publicly available.⁴

In this case, symbol classification means the process of predicting the correct symbol ground truth given its bounding box. Therefore, using the dataset extractor script⁵, the image patches are extracted and stored as individual images with a JSON file describing their annotations.

4.1 Model Architecture and Preprocessing Data

The KuiSCIMA dataset contains 7297 annotated notation instances, from which 133 are excluded from training due to anomalous shapes, as described in Sect. 3.4. Of these, 5168 are simple symbols (meaning that no secondary property is present), and 1996 are composite symbols. Since the dataset is fairly small, the convolutional neural network **FashionCNN** which is taken from [9] is used. It takes input images of size 28×28 .

Regarding preprocessing, each of the image instances is first cleaned up by removing small connected areas which might be printing artifacts. This is done by extracting the contours using OpenCV’s [2] `findContours` function with attributes `cv2.RETR_EXTERNAL` and `cv2.CHAIN_APPROX_NONE`. Areas which have a `contourArea` smaller than 1 are removed. Afterwards, the image is cropped such that white space on the borders is removed.

For training images, additional image augmentations are used. Erosion and dilation are applied with a probability of 10% each, and the image is shrunk such that the longest side is uniformly distributed between 15 and 22, and the aspect ratio is multiplied with a uniformly distributed factor between 0.6 and 1.5. The

⁴ <https://github.com/SuziAI/SuziOMR/tree/main/baseline> (tagged v1.0).

⁵ The dataset extraction procedure is described in https://github.com/SuziAI/gui-tools/blob/main/readme_files/README_ANNOTATION_TOOL.md#extract-omr-dataset-from-corpus (tagged v2.0).

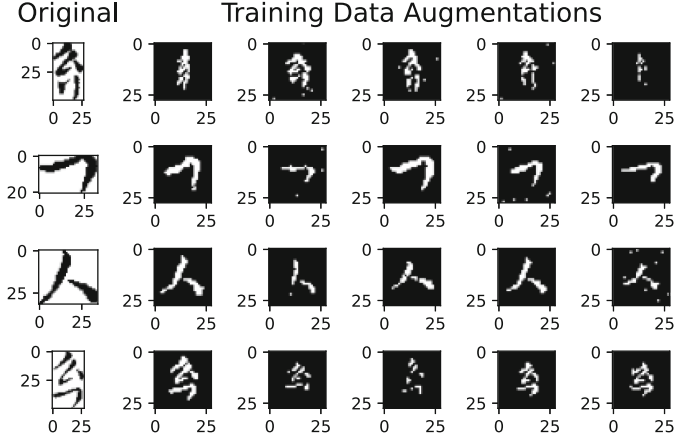


Fig. 6. These data augmentations are used for training the model. The first column contains the image as directly exported from the dataset, and the other column contain some random samples of augmentations of the image in the first column.

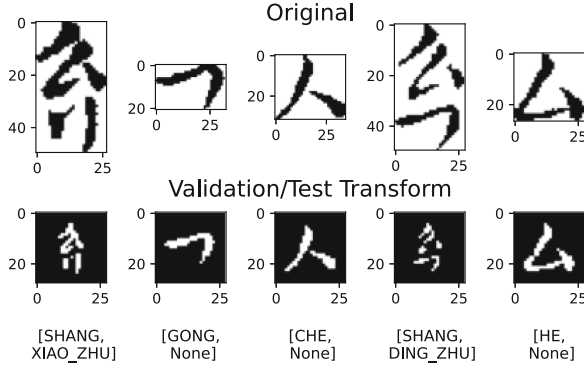


Fig. 7. Here, the transformation of the validation and test data is seen. The first row contains the image as exported from the dataset, the second row contains the corresponding transformation, and the third row contains the annotation.

image is randomly extended by white space to fit into a 28×28 square, salt-and-pepper noise is applied, it is inverted to have black background and white foreground, random rotation between -8 and 8 degrees is applied, and the image is normalized with the test dataset mean of 0.7102 and standard deviation of 0.0914 . The training data augmentations are visualized in Fig. 6. The validation and test images are shrunk such that the longest side has 20 pixels, extended to a square and normalized as described before. This can be seen in Fig. 7.

Each of the pitch and secondary property are predicted using separate instances of **FashionCNN**, with 11 output parameters for pitch, and 7 for secondary (i.e., none resp. the 6 secondary properties present in *suzipu*).

Due to the high label imbalance in the dataset, the training data is augmented to 60000 samples with uniform label frequency. The validation data is also sampled uniformly, but with preservation of the original dataset length.

4.2 Model Selection

A special selection procedure is applied. First, five different model classes are introduced. Each model class consists of a separate model for each pitch and secondary property prediction. They differ in what data is used for training and for validation. The five model classes are:

- The **A11-20%** model is trained on 80% of the KuiSCIMA dataset, while 20% are used as a validation set.
- The **Lu** model is validated on the Lu edition instances, and trained on every other model. Similarly, the **Zhang**, **Siku**, **Zhu** models are validated on the Zhang edition, the *Siku Quanshu* edition, respectively the Zhu edition. Each of those model classes is trained on all data except the validation data and except the Shanghai MS, since it acts as a test set for the model evaluation.

For each of the five model classes, five training trials are made as a random sample of the stochastic training process. In each trial, the model is trained with at most 15 epochs of 600 batches with 100 images each, and the optimizer *Adam* with a learning rate of 0.001 is used. For each trial, the model with the highest accuracy on the validation set is saved. In Fig. 8 (a), the mean trial accuracies and 95% confidence intervals are shown for each model class. In (b), the best trial is tested on the Shanghai MS, consisting of 1440 annotated notation symbols.

4.3 Interpretation

The results show significant differences in the validation set accuracies for each model class. It must again be emphasized that the training and validation stages are performed on uniformly sampled datasets, which is not expected in real-world pieces. The Lu pitch model yields the highest validation accuracy of about 95%, while the **Siku** secondary model yields 93% on their respective validation sets.

From this and the results about variability in [12], one can conclude that both the Lu and *Siku Quanshu* editions do not introduce as much data variability each as the Zhang and Zhu datasets, since high accuracy values are achieved despite dataset exclusion in training and vice versa.

When evaluating on the test data, all models perform with accuracies over 93% on pitch and over 88% on secondary. The **Zhu** pitch model with 95% and the **Siku** secondary model with 94% accuracy perform best. This indicates a high generalization of the models to unseen realistically-distributed data scenarios.

When combining a pitch model and a secondary model together, a mixed model emerges which can predict the complete label of an instance. As a first metric for the *suzipu* notation in the KuiSCIMA dataset, the single CER for the extracted characters is used. The results are found in Fig. 9, and it can be seen that the model performing best is indeed the mixed model from the Zhu pitch and the **Siku** secondary classifier with a CER of 10.4%.

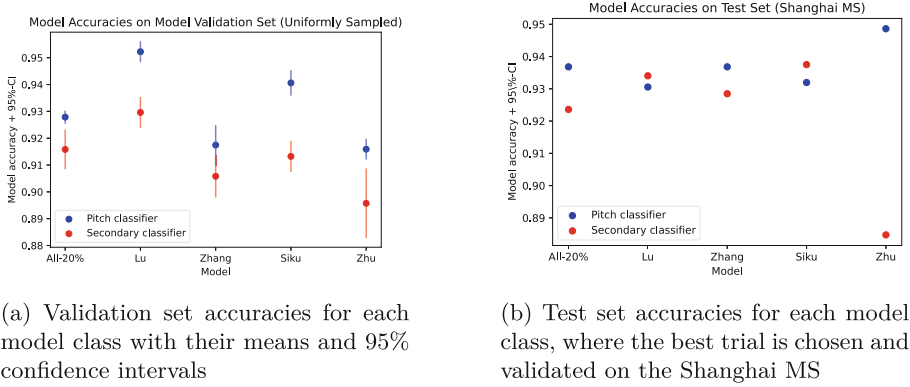


Fig. 8. Validation and test set accuracies for each model class

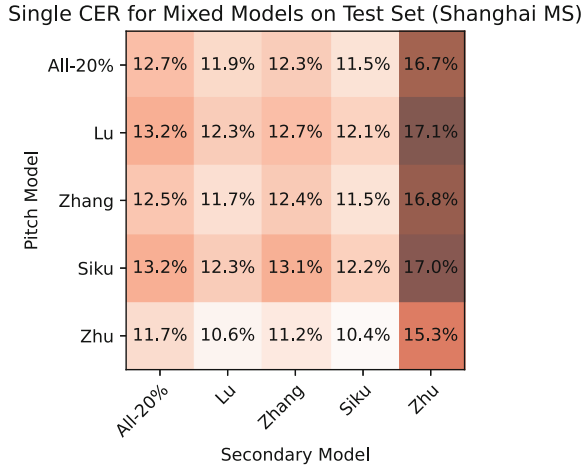


Fig. 9. Here, the single CER are visualized for all mixed models.

5 Conclusion and Future Work

In this paper, KuiSCIMA, the first machine-readable digital OMR dataset containing ancient Chinese *suzipu* notation is introduced. It is publicly available⁶ and based on five different influential editions of Jiang Kui’s 17 *suzipu* pieces in the famous collection *Baishidaoren Geku*. The JSON format employed in this dataset is semantically close to the original notation and features symbol-level annotations of the textual and musical contents with their positions.

KuiSCIMA allows for training and benchmarking OMR models for symbol segmentation and recognition. It is accompanied by an open-source toolbox⁷

⁶ <https://github.com/SuziAI/KuiSCIMA> (tagged v1.0).

⁷ <https://github.com/SuziAI/gui-tools> (tagged v2.0).

which allows for the creation and modification of dataset files, supporting the export MusicXML transnotations. It is able to extract the annotated individual symbol images for direct OMR usage, as demonstrated in baseline experiments on the task of symbol classification for *suzipu* notation in Sect. 4.

The structure and contents of the KuiSCIMA dataset allow for musicological uses, e.g., the computer-aided comparison of the different editions, music analysis based on n-grams, and other statistics-based methods.

In the future, we propose new research based on KuiSCIMA as follows:

- The re-encoding of KuiSCIMA into a standard notation format like the Music Encoding Initiative (MEI⁸).
- The digitization of other *suzipu*-based music, e.g., *Xi'an Guyue* 西安鼓乐.
- Generalizing KuiSCIMA to incorporate all pieces of *Baishidaoren Gequ* into the dataset, including ten *lilüpu* and the *jianzipu* piece *Guyuan* 古怨.
- The development of end-to-end OMR systems for *suzipu* notation.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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